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(54) STEERING APPARATUS FOR VEHICLE

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(57) ABSTRACT

A steering apparatus for a vehicle including a housing, a worm wheel secured inside the housing and mounted on a steering shaft, a worm shaft operably engaged with the worm wheel to rotate, a bearing mounted on the worm shaft, and a damper rotatably surrounds the bearing and elastically presses the bearing to allow the worm shaft to engage with the worm wheel.

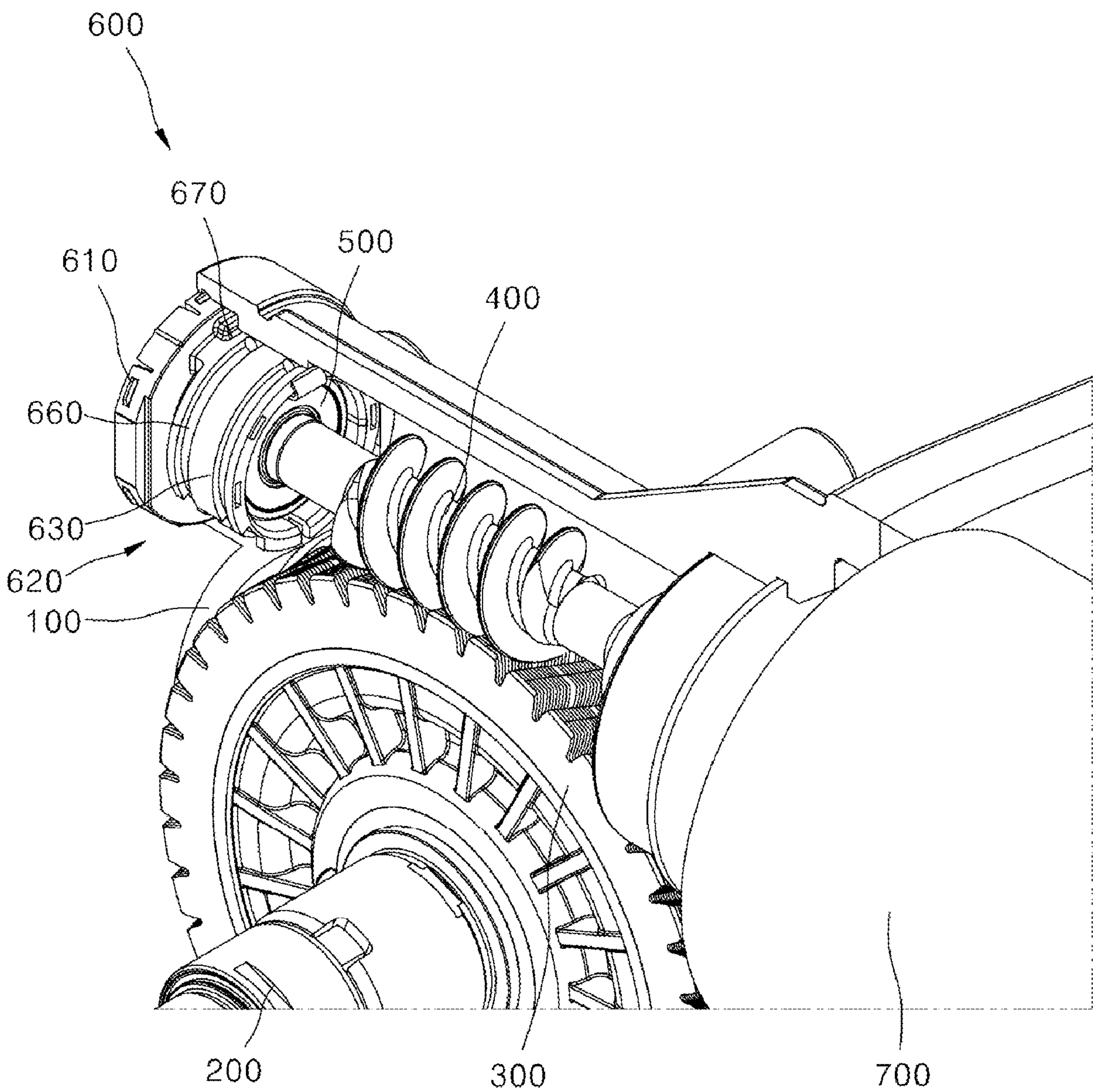


FIG. 1

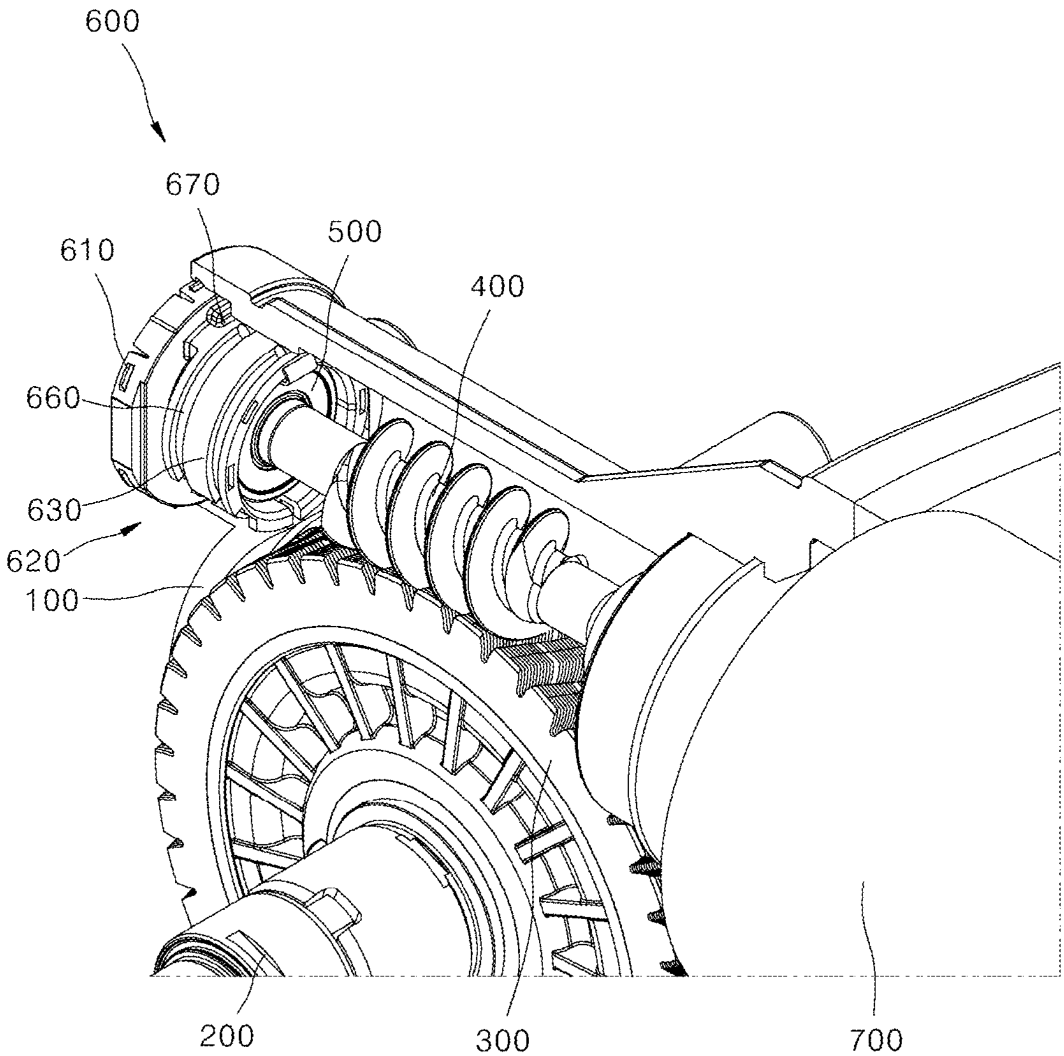


FIG. 2

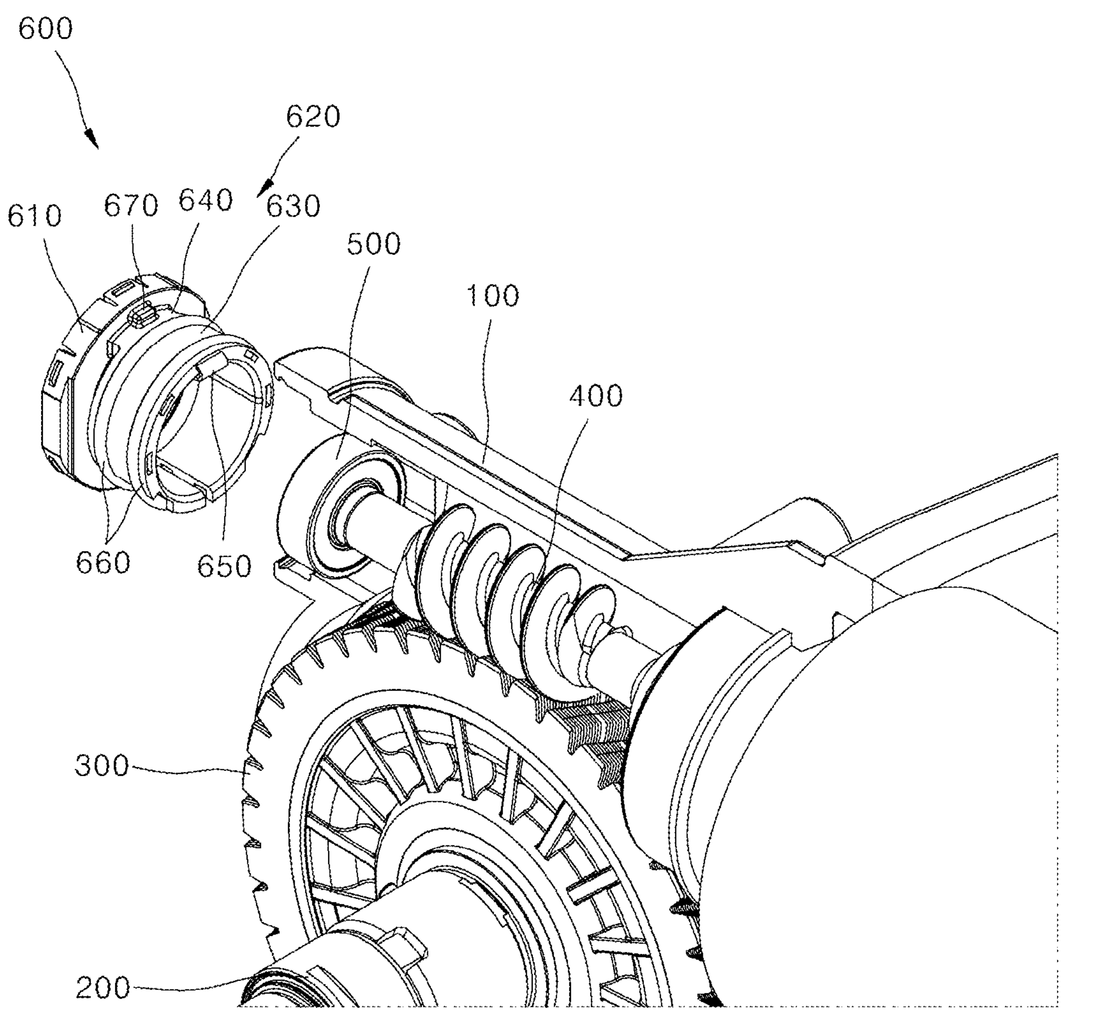


FIG. 3

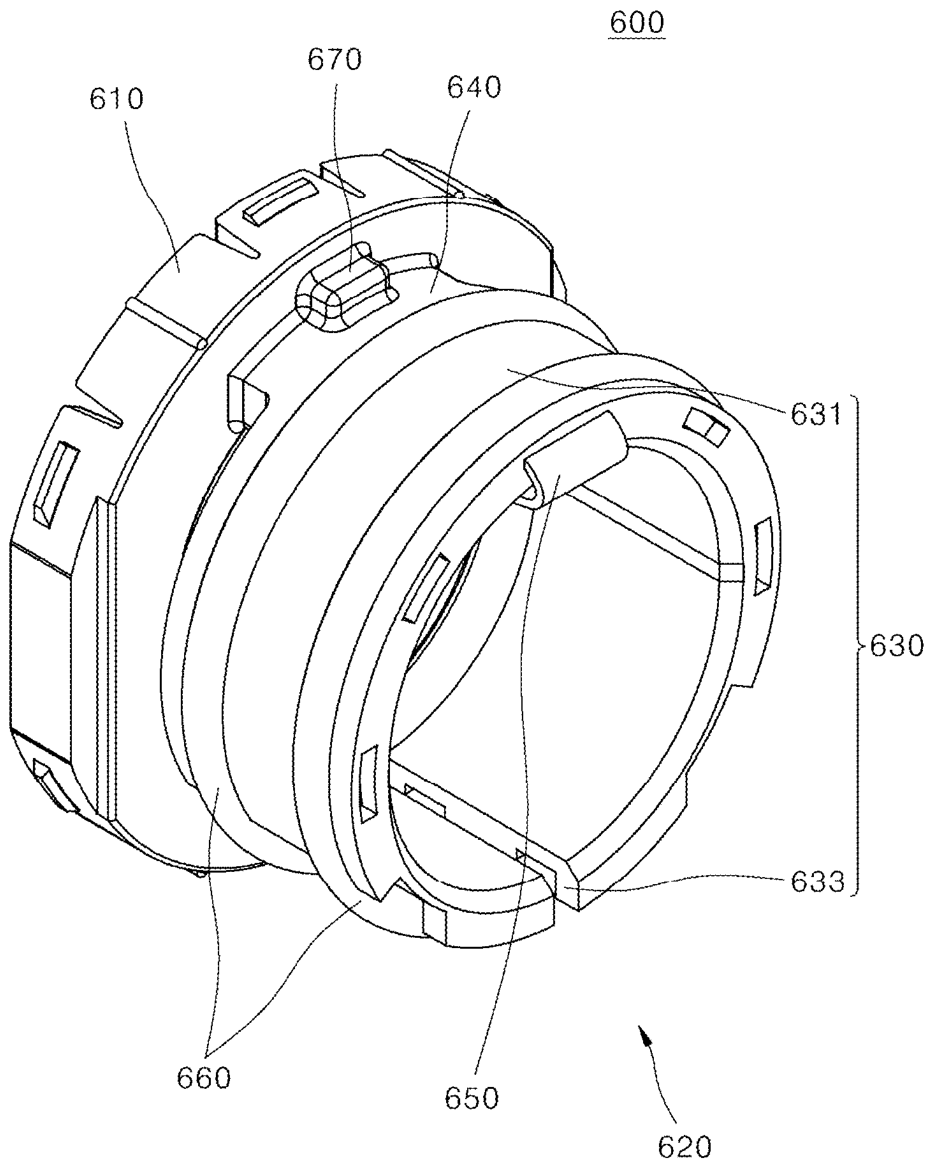


FIG. 4

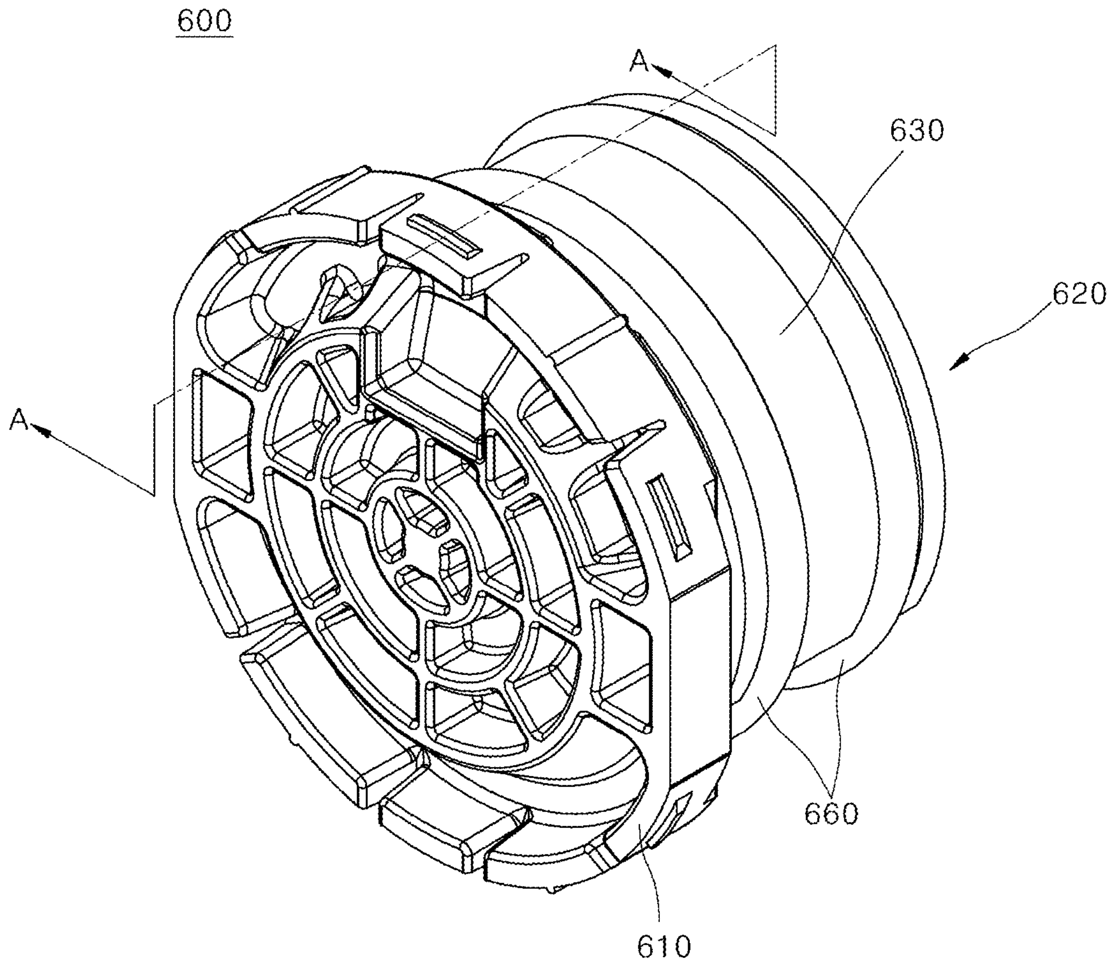


FIG. 5

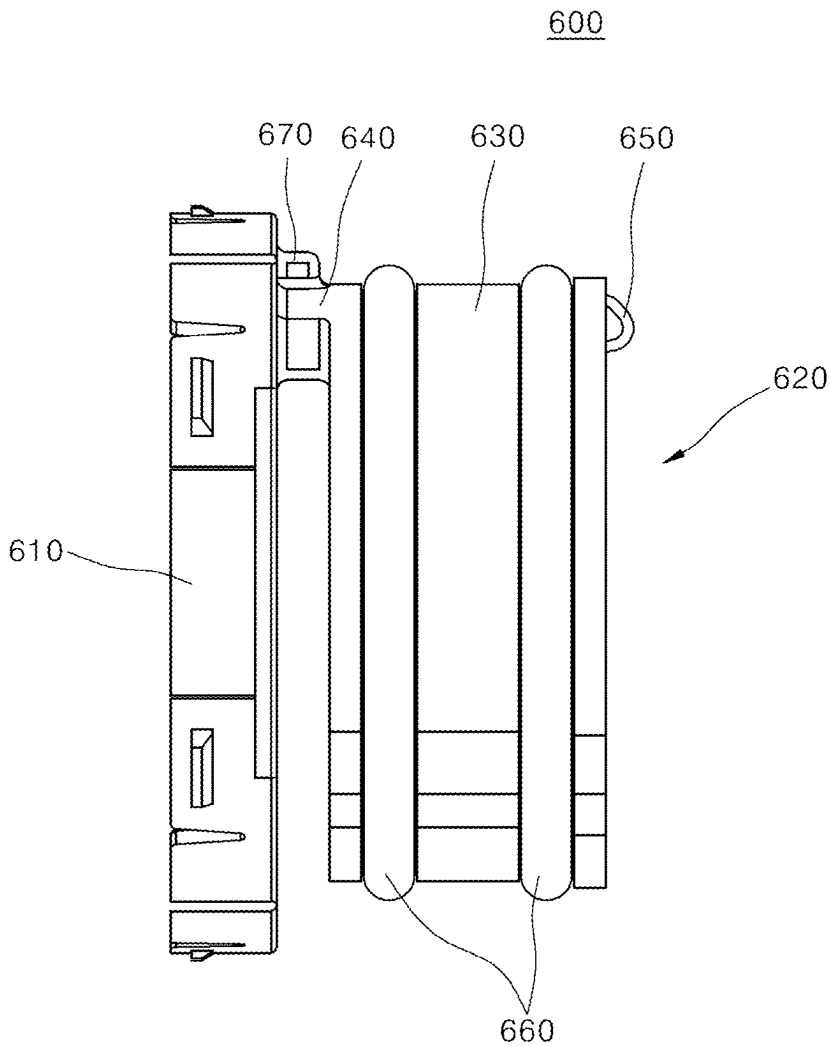


FIG. 6

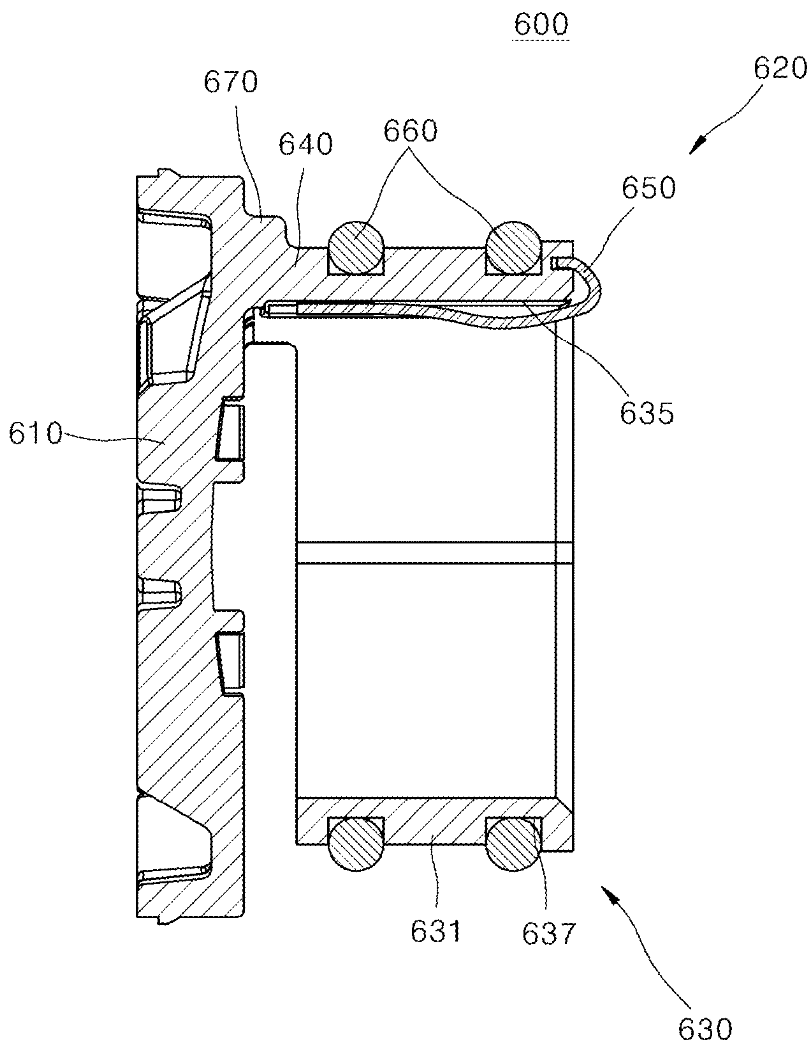


FIG. 7

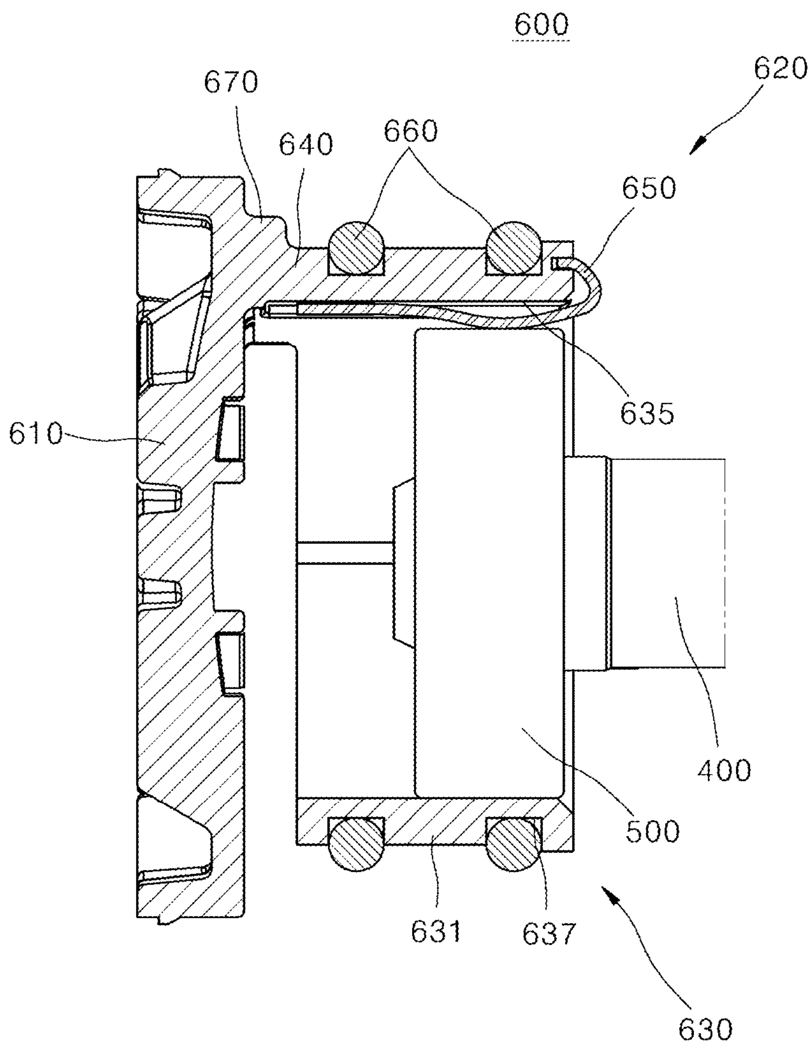


FIG. 8

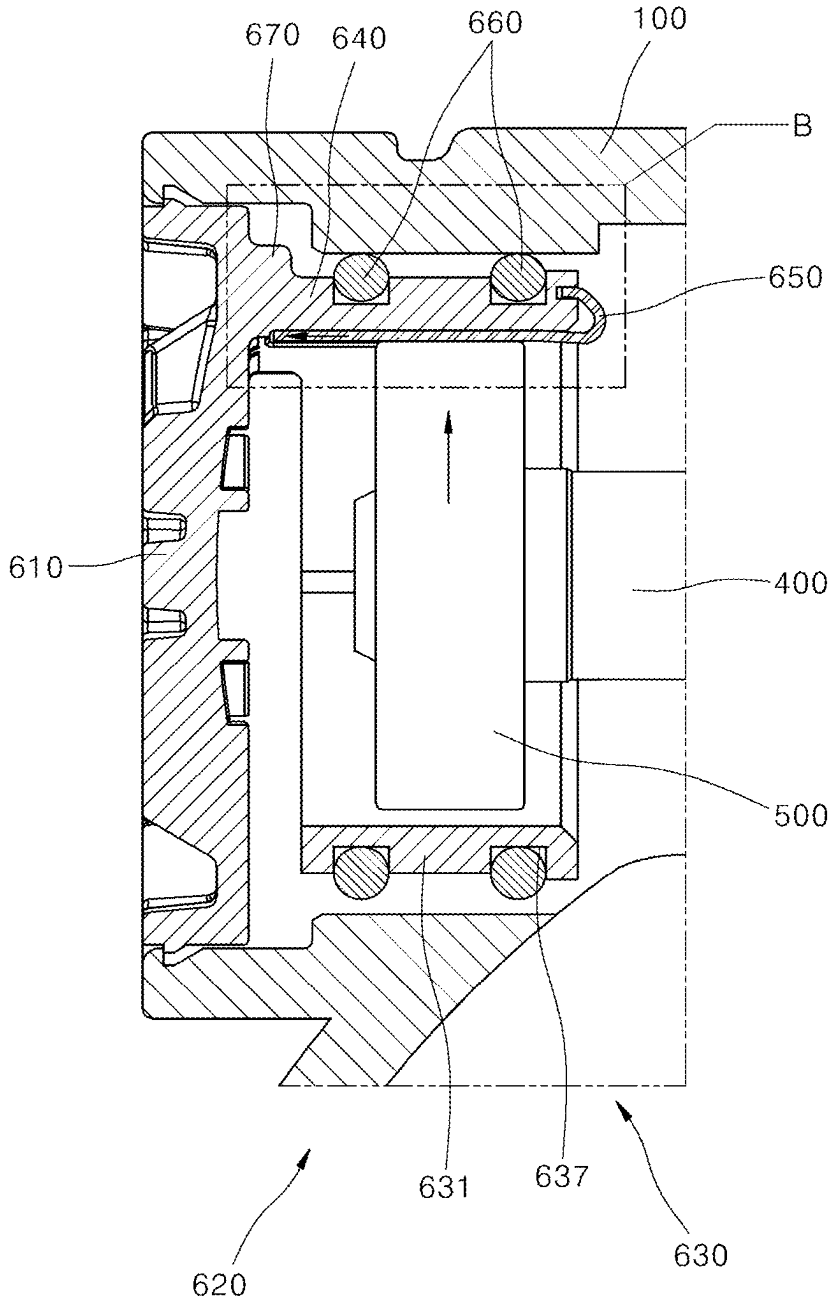
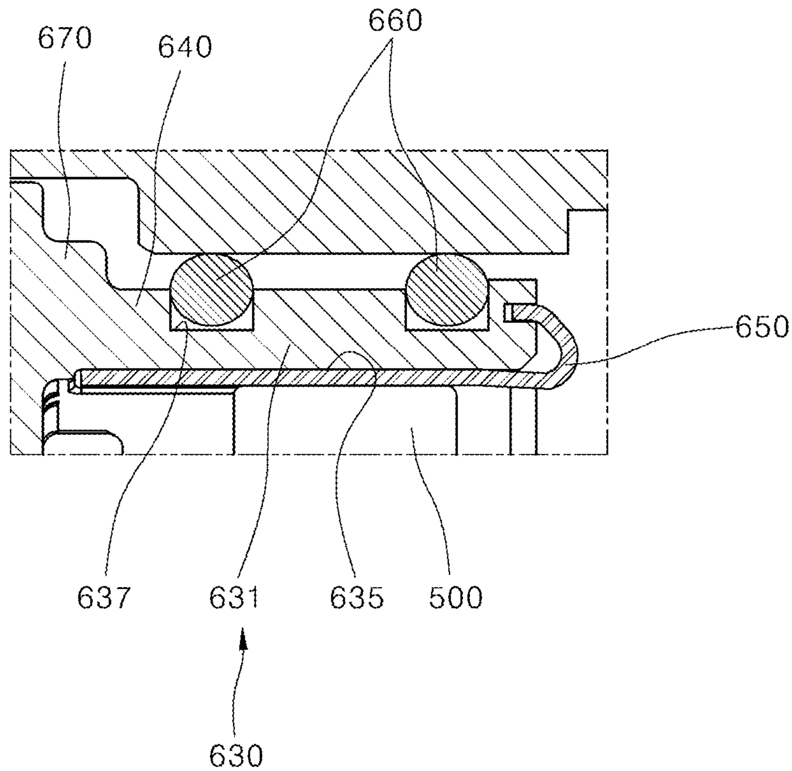


FIG. 9



STEERING APPARATUS FOR VEHICLE

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority from and the benefit under 35 U.S.C. § 119 of Korean Patent Application No. 10-2024-0028255, filed on Feb. 27, 2024, in the Korean Intellectual Property Office, the entire disclosure of which is incorporated by reference for all purposes.

BACKGROUND

Technical Field

[0002] Exemplary embodiments of the present disclosure relate to a steering apparatus for a vehicle and, more particularly, to a steering apparatus for a vehicle capable of preventing tilting and noise of a worm shaft.

Description of the Related Art

[0003] In a conventional steering apparatus, components such as a spring, sliding damper, and end cover assembly are configured in a lower bearing to compensate for the rotation and tilting of a worm shaft when the worm shaft and worm wheel are driven. The axial distance between the worm wheel and the worm shaft is not consistently maintained, causing the worm shaft to tilt depending on a pressure angle as the worm shaft rotates. The tilting of the worm shaft increases rattle noise and friction, and also causes the generation of abnormal noise when the worm shaft tilts more than a desired tilt amount. Therefore, these issues require improvement.

[0004] The related art of the present disclosure is disclosed in Korean Patent No. 10-2261787 (registered on Jun. 1, 2021 and entitled "STEERING APPARATUS FOR VEHICLE").

SUMMARY

[0005] Various embodiments of the present disclosure are directed to a steering apparatus for a vehicle capable of preventing tilting and noise of a worm shaft.

[0006] A steering apparatus for a vehicle according to the present disclosure includes: a housing, a worm wheel configured to be housed inside the housing and be mounted on a steering shaft, a worm shaft configured to be engaged with the worm wheel to rotate; a bearing configured to be mounted on the worm shaft, and a damper configured to rotatably surround the bearing and elastically press the bearing to allow the worm shaft to engage with the worm wheel.

[0007] In an embodiment, the damper may include a cover configured to close the open housing and an elastic part configured to extend from the cover, elastically press the bearing, and allow the worm shaft to engage with the worm wheel.

[0008] In an embodiment, the elastic part may include an elastic body configured to be spaced apart from the cover and surround the bearing, a connection configured to connect the cover and the elastic body, and an elastic arm configured to be mounted on the elastic body and provide an elastic force to the bearing while being elastically deformed upon contact with the bearing.

[0009] In an embodiment, the connection may connect the elastic body to the cover in a cantilever shape.

[0010] In an embodiment, the elastic body may include an elastic body guide configured to be spaced apart from the cover and surround the bearing and an elastic cutout configured to be cut into the elastic body guide and allow the elastic body guide to be elastically deformed when in contact with the bearing.

[0011] In an embodiment, the elastic body may further include an elastic movement path configured to be formed to pass through an inner surface of the elastic body guide and provide a path for the bearing elasticity providing part to move when in contact with the bearing.

[0012] In an embodiment, the elastic part may further include an O-ring configured to be mounted on an outer surface of the elastic body and be elastically deformed when in contact with an inner surface of the housing.

[0013] In an embodiment, a plurality of the O-rings may be mounted along the longitudinal direction of the elastic body.

[0014] In an embodiment, the elastic body may include an O-ring mount groove concave along a circumferential direction of the elastic body, and the O-ring may be mounted in the O-ring mount groove.

[0015] In an embodiment, the elastic part may further include a reinforcing part configured to connect the cover and the connection.

[0016] With the steering apparatus for a vehicle of the present disclosure, the tilting and noise of the worm shaft may be prevented by the damper.

[0017] In addition, with the steering apparatus for a vehicle, the tilting of the worm shaft is controlled by a single damper, which reduces the number of components and simplifies an assembly process, thereby reducing manufacturing costs and improving productivity.

BRIEF DESCRIPTION OF THE DRAWINGS

[0018] FIG. **1** is a schematic perspective view of a steering apparatus for a vehicle according to an embodiment of the present disclosure.

[0019] FIG. **2** is a schematic assembled perspective view of a steering apparatus for a vehicle according to an embodiment of the present disclosure.

[0020] FIG. **3** is a schematic perspective view of a damper from one direction according to an embodiment of the present disclosure.

[0021] FIG. **4** is a schematic perspective view of a damper from the other direction according to an embodiment of the present disclosure.

[0022] FIG. **5** is a schematic side view of a damper according to an embodiment of the present disclosure.

[0023] FIG. **6** is a schematic side sectional view taken along line A-A of FIG. **5**.

[0024] FIG. **7** is a schematic side sectional view showing a state before an operation of a bearing and a worm shaft mounted in a damper of a steering apparatus for a vehicle according to an embodiment of the present disclosure.

[0025] FIG. **8** is a schematic side sectional view showing an operation of a bearing and a worm shaft mounted in a damper of a steering apparatus for a vehicle according to an embodiment of the present disclosure.

[0026] FIG. **9** is an enlarged partial side sectional view schematically showing B of FIG. **8**.

DETAILED DESCRIPTION

[0027] Embodiments of a steering apparatus for a vehicle according to the present disclosure will be described hereinafter with reference to the accompanying drawings. For clarity and convenience in description, thicknesses of lines, sizes of constituent elements, and the like may be illustrated in exaggerated proportions in the drawings.

[0028] In addition, the terms used below are defined in consideration of the functions thereof in the present disclosure and may vary depending on the intention of a user or an operator or common practice. Therefore, these terms should be contextually defined in light of the present specification.

[0029] FIG. **1** is a schematic perspective view of a steering apparatus for a vehicle according to an embodiment of the present disclosure. FIG. **2** is a schematic assembled perspective view of a steering apparatus for a vehicle according to an embodiment of the present disclosure. FIG. **3** is a schematic perspective view of a damper from one direction according to an embodiment of the present disclosure. FIG. **4** is a schematic perspective view of a damper from the other direction according to an embodiment of the present disclosure. FIG. **5** is a schematic side view of a damper according to an embodiment of the present disclosure. FIG. **6** is a schematic side sectional view taken along line A-A of FIG. **5**. FIG. **7** is a schematic side sectional view showing a state before operation of a bearing and a worm shaft mounted in a damper of a steering apparatus for a vehicle according to an embodiment of the present disclosure. FIG. **8** is a schematic side sectional view showing an operation of a bearing and a worm shaft mounted in a damper of a steering apparatus for a vehicle according to an embodiment of the present disclosure. FIG. **9** is an enlarged partial side sectional view schematically showing B of FIG. **8**.

[0030] Referring to FIGS. **1** to **9**, a steering apparatus for a vehicle according to an embodiment of the present disclosure may include a housing **100**, a steering shaft **200**, a worm wheel **300**, a worm shaft **400**, a bearing **500**, and a damper **600**.

[0031] Inside the housing **100**, the steering shaft **200**, the worm wheel **300**, the worm shaft **400**, the bearing **500**, and the damper **600** may be mounted. The housing **100** may be formed with one side (left side in FIGS. **1** and **2**) open, and the damper **600** may be mounted to the open housing **100**.

[0032] The steering shaft **200** may be connected to a steering wheel of a vehicle. The worm wheel **300** may be housed inside the housing **100** and be mounted on the steering shaft **200**.

[0033] The worm shaft **400** may be engaged with the worm wheel **300** to rotate. The worm shaft **400**, engaged with the worm wheel **300** to rotate, may be axially connected to a motor **700**. The motor **700** may be connected to one side (right side in FIG. **1**) of the worm shaft **400**, and a ring-shaped bearing **500** may be mounted on the other side (left side in FIG. **1**) of the worm shaft **400**.

[0034] The bearing **500** may be mounted on an end (left end in FIG. **1**) of the worm shaft **400**. The bearing **500** may be mounted on the end of the worm shaft **400** to rotatably support the worm shaft **400** and reduce friction generated by the rotation of the worm shaft **400**.

[0035] The damper **600** may rotatably surround the bearing **500** and elastically press the bearing **500** to allow the worm shaft **400** to engage with the worm wheel **300**. The damper **600** may elastically press the bearing **500** to direct the worm shaft **400** toward the worm wheel **300**, thereby preventing or reducing tilting and noise of the worm shaft **400**.

[0036] The damper **600** may include a cover **610** and an elastic part **620**. The cover **610** may close one side (left side in FIG. **1**) of the open housing **100**. The cover **610** may close the open housing **100** to prevent the entry of foreign substances into the housing **100**.

[0037] The elastic part **620** may extend from the cover **610**, elastically press the bearing **500**, and allow the worm shaft **400** to engage with the worm wheel **300**. The elastic part **620** may include an elastic body **630**, a connection **640**, and an elastic arm **650**.

[0038] The elastic body **630** may be spaced apart from the cover **610** and surround the bearing **500**. The elastic body **630** may be formed in a ring shape to surround the bearing **500**.

[0039] The elastic body **630** may include an elastic body guide **631** and an elastic cutout **633**. The elastic body guide **631** may be spaced apart from the cover **610** and surround the bearing **500**. The elastic body guide **631** may be formed in a ring shape to surround the bearing **500**. The elastic body guide **631** may be formed with one side (lower side in FIG. **3**) open.

[0040] The elastic cutout **633** may be cut into one side (lower side in FIG. **3**) of the elastic body guide **631** and allow the elastic body guide **631** to be elastically deformed when in contact with the bearing **500**.

[0041] When assembled into the housing **100**, the elastic part **620** may absorb lateral clearance with the bearing **500** as the elastic body guide **631** contracts laterally due to the elastic cutout **633**.

[0042] After contraction, an internal shape of the elastic body guide **631** may maintain an elliptical shape, thereby preventing or reducing an up-and-down tilt of the bearing **500**.

[0043] The elastic body **630** may further include an elastic movement path **635**. The elastic movement path **635** may be formed to pass through an inner surface of the elastic body guide **631** and provide a path for the bearing elasticity providing part **650** to move when in contact with the bearing **500**. The elastic movement path **635** may provide a path along which one end (left end in FIG. **6**) of the bearing elasticity providing part **650** may move, and prevent the bearing elasticity providing part **650** from disengaging. The elastic movement path **635** may be formed in a size that allows the bearing elasticity providing part **650** to move.

[0044] The elastic body **630** may further include an O-ring mount groove **637**. The O-ring mount groove **637** may be concave along the circumferential direction of the elastic body guide **631**. An O-ring **660** may be mounted in the O-ring mount groove **637**. The O-ring **660** may be mounted in the O-ring mount groove **637**, thereby preventing the O-ring **660** from disengaging. The number of O-ring mount grooves **637** may match the number of O-rings **660**.

[0045] The connection **640** may connect the cover **610** and the elastic body **630**. The connection **640** may be connected to a portion of the cover **610** on one side (left side in FIG. **5**) and to a portion of the elastic body **630** on the other side (right side in FIG. **5**). The connection **640** may connect the elastic body **630** to the cover **610** in a cantilever shape. The elastic body guide **631** of the elastic body **630** may be connected to the cover **610** in a cantilever shape by the connection **640**, and may be elastically deformed by the bearing **500** tilting up and down.

[0046] The bearing elasticity providing part **650** may be mounted on the elastic body **630** and provide an elastic force (elastic restoring force) to the bearing **500** while being elastically deformed upon contact with the bearing **500**. One side (right side in FIG. **6**) of the bearing elasticity providing part **650** may be fixed to the elastic body guide **631** of the elastic body **630**. The other side (left side in FIG. **6**) of the bearing elasticity providing part **650** may be movably inserted into the elastic movement path **635** of the elastic body **630**. A portion of the surface (lower surface in FIG. **6**) of the bearing elasticity providing part **650** may be exposed from the elastic movement path **635** of the elastic body **630** and come into contact with the bearing **500**.

[0047] The bearing elasticity providing part **650** may be elastically deformed when in contact with the bearing **500** that moves due to the worm shaft **400** tilting up and down. The exposed lower surface of the bearing elasticity providing part **650** may be compressed in contact with the bearing **500**, and the other side of the bearing elasticity providing part **650** may be moved in the elastic movement path **635**. The bearing elasticity providing part **650** may be made of a leaf spring.

[0048] The extent to which the bearing elasticity providing part **650** is exposed in the elastic movement path **635** of the elastic body **630** may be set depending on the amount of tilting of the bearing **500** and the worm shaft **400**.

[0049] The elastic part **620** may further include the O-ring **660**. The O-ring **660** may be mounted on an outer surface of the elastic body **630** and elastically deformed when in contact with an inner surface of the housing **100**. The O-ring **660** may be formed in a ring shape and be mounted in the O-ring mount groove **637** of the elastic body **630**.

[0050] The O-ring **660** may be elastically deformed when in contact with the inner surface of the housing **100**. The O-ring **660** may prevent the entry of foreign substances and the generation of noise while being elastically deformed upon contact with the inner surface of the housing **100**.

[0051] The O-ring **660** may be made of an elastically deformable material, including rubber, silicone, urethane, or the like.

[0052] A plurality of the O-rings **660** may be mounted along the longitudinal direction (right-left direction in FIGS. **5** and **6**) of the elastic body **630**. A plurality of the O-rings **660** may be mounted along the longitudinal direction of the elastic body guide **631** of the elastic body **630**, thereby preventing the entry of foreign substances and the generation of noise.

[0053] The elastic part **620** may further include a reinforcing part **670**. The reinforcing part **670** may connect the cover **610** and the connection **640**. The reinforcing part **670** may connect the cover **610** and the connection **640** so that the connection between the cover **610** supported by the connection **640** and the elastic body **630** of the elastic part **620** may be stably maintained. The thickness of the reinforcing part **670** may be adjusted depending on an elastic force of the connection **640**.

[0054] Referring to FIGS. **7** to **9**, an operation of a steering apparatus for a vehicle according to an embodiment of the present disclosure may be described.

[0055] In FIG. **7**, the bearing elasticity providing part **650** remains in contact with the bearing **500**.

[0056] Referring to FIGS. **8** and **9**, the bearing **500** mounted on the worm shaft **400** may be moved upward by the tilting worm shaft **400**.

[0057] The bearing elasticity providing part **650** may be elastically deformed when in contact with the bearing **500** that moves due to the worm shaft **400** tilting up and down.

[0058] The bearing elasticity providing part **650** may provide an elastic force (elastic restoring force) to the bearing **500** while being elastically deformed upon contact with the bearing **500**. The exposed lower surface of the bearing elasticity providing part **650** may be compressed in contact with the bearing **500**, and the other side (left side in FIG. **8**) of the bearing elasticity providing part **650** may be moved in the elastic movement path **635**.

[0059] The bearing **500** may move downward due to the elastic restoring force of the bearing elasticity providing part **650**, and the worm shaft **400** and the worm wheel **300** may maintain engagement, thereby preventing tilting and noise.

[0060] With the steering apparatus for a vehicle of the present disclosure, the tilting of the worm shaft **400** may be easily managed by the damper **600**.

[0061] In addition, with the steering apparatus for a vehicle, the tilting of the worm shaft **400** may be controlled by a single damper **600**, which reduces the number of components and simplifies an assembly process, thereby reducing manufacturing costs and improving productivity.

[0062] Although specific embodiments of the present disclosure have been described above, the spirit and scope of the present disclosure are not limited to these specific embodiments. Various modifications and variations are possible within the scope of not changing the gist of the present disclosure as stated in the claims by those skilled in the art to which the present disclosure pertains.

What is claimed is:

1. A steering apparatus for a vehicle comprising:

a housing;

a worm wheel is positioned inside the housing and be mounted on a steering shaft;

a worm shaft operably engaged with the worm wheel to rotate;

a bearing mounted on the worm shaft; and

a damper rotatably surrounds the bearing and elastically presses the bearing to allow the worm shaft to engage with the worm wheel.

2. The steering apparatus of claim **1**, wherein the damper comprises:

a cover that is mounted on an opening of the housing; and

an elastic part is protruded from the cover, elastically presses the bearing, and allows the worm shaft to engage with the worm wheel.

3. The steering apparatus of claim **2**, wherein the elastic part comprises:

an elastic body spaced apart from the cover and surrounds the bearing;

a connection to connect the cover and the elastic body; and

an elastic arm mounted on the elastic body and providing an elastic force to the bearing while being elastically deformed upon contact with the bearing.

4. The steering apparatus of claim **3**, wherein the connection connects the elastic body to the cover in a cantilever shape.

5. The steering apparatus of claim **3**, wherein the elastic body comprises:

an elastic body guide spaced apart from the cover and surrounds the bearing; and

an elastic cutout is cut into the elastic body guide and allows the elastic body guide to be elastically deformed when in contact with the bearing.

6. The steering apparatus of claim **5**, wherein the elastic body further comprises an elastic movement path formed to pass through an inner surface of the elastic body guide and provides a path for the elastic arm to move when in contact with the bearing.

7. The steering apparatus of claim **3**, wherein a plurality of O-rings are mounted along the longitudinal direction of the elastic body.

8. The steering apparatus of claim **7**, wherein the elastic part further comprises an O-ring mounted on an outer surface of the elastic body and is elastically deformed when in contact with an inner surface of the housing.

9. The steering apparatus of claim **7**, wherein:

the elastic body comprises an O-ring mount groove positioned along a circumferential direction of the elastic body, and

the O-ring is mounted in the O-ring mount groove.

10. The steering apparatus of claim **3**, wherein the elastic part further comprises a reinforcing part to connect the cover and the connection.

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